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REVIEW PAPER

Deep eutectic solvents in biomass pretreatment for green approach: a comprehensive review

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Abstract

The increasing demand for efficient and environmentally friendly biomass processing techniques has led to interest in utilizing deep eutectic solvents (DESs) as a "green" alternative to conventional pretreatment reagents. The current review critically examines the utilization of DESs for pretreatment of lignocellulosic biomass and biorefinery operations, recyclability, composition, and solvent capability. Key outcomes show that DESs offer superior lignin removal, hemicellulose solubilization, and enzymatic hydrolysis enhancement, leading to good yields of biofuels, sugars, and other value-added bioproducts. The review brings together knowledge from life cycle studies and techno-economic studies and addresses current limitations of high viscosity and non-scalability, and proposes how these can be avoided. Through the incorporation of recent advances, this study establishes the platform of DES-based biomass conversion and offers a roadmap for the future creation of efficient, scalable, and sustainable biorefinery systems.

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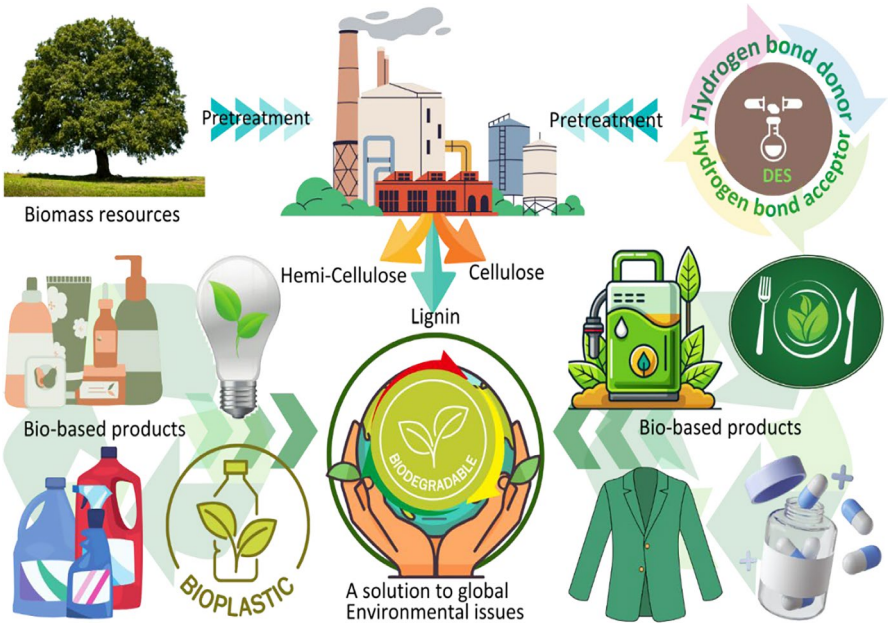
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Graphical abstract



Keywords Lignocellulose · Extraction · Ionic liquid · Biorefinery · Life cycle assessment

Introduction

Lignocellulosic biomass can be broadly categorized into three main groups: energy crops, waste biomass, and virgin biomass [1]. Virgin biomass means all naturally existing terrestrial vegetation, including trees, shrubs, and grasses [2]. Conversely, waste biomass is a low-value byproduct generated by various industries, including the forestry sector and agriculture. Energy crops, like switchgrass and elephant grass, are cultivated specifically for their high yields of lignocellulosic material, which can be utilized in the production of second-generation biofuels [3]. Lignocellulosic biomass is characterized by a broad array of physical attributes and mechanical strengths, with lignin (a complex natural polymer/rigid component), cellulose (linear homopolymer/fiber maker), and hemicelluloses being almost all universal components. Cellulose and hemicelluloses make up almost 70% of the biomass composition; these are bonded to lignin by hydrogen bonds, which in some cases are chemically covalent bonds [4]. This structural configuration gives strong resistance of lignocellulosic biomass to degradation. Cellulose, in general, forms most of the skeletal backbone and is encumbered by hemicelluloses and lignin [5].

Combustion degrees for toxic air pollutants from agricultural residues are on the rise due to their illegal disposal as street garbage or by open burning, while the rays of resource depletion shine brighter as nonrenewables reach the end of their expected lives or are simply consumed [6–8]. Lignocellulosic biorefinery refers to the green transformation of lignocellulosic biomass consisting of cellulose, hemicellulose, and lignin into energy, chemicals, fuels, and high-value bio-based materials [9]. It is modeled after the petroleum refinery model but using renewable biomass instead of fossil fuel, in line with BCG principles. Various sugarcane by-products, such as straw, bagasse, and molasses, can be used to produce items like ethanol, butanol, xylitol, biogas, polyhydroxyalkanoates, and nanocellulose through biochemical, thermochemical, and enzymatic routes in second-generation (2G) biorefineries [10]. These processes not only conserve waste but also generate new sources of revenue, hence promoting circularity and sustainability in the biomass value chain [11]. Lignocellulosic biomass is a recyclable and plentiful resource that holds great potential to increase the production of green fuels and chemicals. Global production of biomass is 181.5 billion tons, with only 8.2 billion suitable for utilization in 2022 [12]. The lignocellulosic biomass market is projected to expand considerably, with its value anticipated to rise from $4.3 billion in 2024 to $9.1 billion by 2034, underscoring the growing recognition of its economic and environmental advantages [13]. Lignocellulosic biomass primarily consists of cellulose, hemicellulose, and lignin [14, 15]. The rigid and complex structure of lignocellulose makes it less advantageous for applications [16]. The biomass pretreatment process is critical for converting biomass into high-value products through the breakdown of lignocellulosic structures, leading to easy accessibility for subsequent processing [17].

The separation and purification of the lignocellulosic components allow huge opportunities for value addition and improvement in plant biomass waste utilization. Presently, biological, hydrothermal, thermochemical, and chemical processes, among others, have been used as pretreatment methods [18, 19]. In biomass fractionation for lignocellulosic pretreatment, various chemical methods have been offered, i.e., ionic liquids and organic solvents [20] [21] [22]. Low-toxic ionic liquids [23, 24], primarily composed of ions with melting points below 100 °C, are often favored over conventional volatile solvents [25–27]. However, although they are highly effective at separating lignocellulosic materials, their high cost and potential environmental impact limit their use in large-scale sustainable production processes. Consequently, the search for a cost-efficient, nontoxic, and environmentally safe solvent is necessary. DESs are increasingly being touted as green or eco-friendly solvents due to a range of environmentally friendly characteristics that distinguish them from conventional chemical solvents and even some ionic liquids. DESs are usually low-cost, biodegradable mixtures of chemicals such as choline chloride, glycerol, organic acids, or urea that are commonly regarded as being safe and nontoxic [28, 29]. Their inherent inertness and low volatility make them noncombustible and minimize air pollution hazards. Unlike traditional solvents that typically must be thoroughly purified and create hazardous by-products, DESs are easy to prepare under mild conditions without the need for complex synthesis or purification processes. Besides, they possess high recyclability, with many DES formulations displaying stable activity even after multiple cycles of reuse. Their low environmental

persistence and biodegradability also enhance their sustainability profile. In addition to these environmentally desirable features, DESs are also highly soluble and selective in solvating lignin, so biomass fractionation can be carried out efficiently with no formation of fermentation inhibitors [30]. This combination of safety, efficiency, and environmental acceptability makes DESs a viable candidate for sustainable biorefinery processes [31]. Various traditional processes have been adopted for the pretreatment of biomass in industrial applications, particularly in larger usages. However, the processes do cause pollution to the environment and introduce much greater difficulties for large-scale recovery, while identification of an alternative pretreatment using a deep eutectic solvent appears to have many advantages, including biodegradability, good stability, environmental friendliness, and recyclability [32, 33].

DESs are formed by the combination of two or more components that possess strong hydrogen bond acceptor and donor properties [34], giving rise to a mixture with a lower melting point than its components [35]. Their interaction with the biomass can most probably be explained by an acid–base catalytic mechanism [36]. During this process, the catalytic action by hydrogen protons leads to the breaking of weak bonds within the ether-alkyl groups connecting the phenyl propane units of lignin along with the disruption of some hydrogen bonds and covalent bonds e.g., benzyl ether (–O–CH_2–), benzyl ester (–$COOCH_2$–), and phenyl glycosidic bonds that are formed between hemicellulose and lignin and render the biomass recalcitrant to biodegradation [37]. This trait renders DESs quite attractive as an alternative for biomass processing since they offer unique solvation characteristics and effectively rupture complex lignocellulosic structures [38, 39]. Additionally, several DES systems of great promise have been discussed in the latest studies [40–43].

DESs are efficient in the solubilization of the lignin and hemicellulose components of the biomass matrix to produce more porous, cellulose-rich material, thus easier to degrade [44]. Besides dissolving lignocellulose, DESs also enhance the efficiency of various pretreatment methods. They can act as cosolvents with conventional ethanol or water to further solubilize lignin, reaching cellulose accessibility. In addition, DESs act as catalysts that help in the depolymerization of lignin and the hydrolysis of cellulose, thus increasing useful product yields [45]. While DESs are indeed promising in the pretreatment of biomass, there are still some critical issues that need addressing. These include the high cost of DESs, the possibility of toxic components, and scant information available regarding their long-term environmental effects. However, it is clear that the application of DESs particularly in lignocellulose pretreatment provides ample opportunity for the development of low-cost and benign bioproducts.

Some reviews have already covered the use of DESs in the processing of biomass. For example, Tang et al. [41] were among the first to report green processing of lignocellulosic biomass with DESs, while Bjelić et al. [42] and Bao et al. [43] covered the application of DESs in lignin separation, catalysis, and valorization. However, all these reviews are focused on DES composition or a specific case study and lack an integrated and interdisciplinary overview. The present study fills this lacuna by comprehensively evaluating the forms, physicochemical properties, and functional activity of DESs in biomass pretreatment and combining data on lignin,

hemicellulose, and cellulose extraction with recent developments in enzymatic hydrolysis, sugar recovery, and biofuel production. Unlike other reviews, the present study also considers sustainability indices, such as life cycle assessment (LCA) and techno-economic analysis (TEA), to analyze the environmental and economic viability of DES-based technologies. The review has been structured to follow the whole biomass valorization process, from fractionation and design of solvents for biomass to recovery of products, recycling of DES, and system sustainability. These observations not only highlight the advantages of DESs, such as low toxicity, recyclability, and biodegradability, but also expose critical challenges such as co-extraction effects, high viscosity, and scalability. Collectively, this paper provides an extensive and forward-looking manual for the design of DES-facilitated biorefineries as part of an effective circular bioeconomy.

Deep eutectic solvents-based lignocellulosic pretreatment technology

Deep eutectic solvents and their types

DESs represent a new class of environmentally friendly [46] solvents formed by combining two or more components that interact through hydrogen bonding. These solvents show great potential for biomass pretreatment [47]. DESs have similar characteristics like ionic liquids, such as low volatility and tunable polarity produced as a eutectic mixture of a salt combined with a hydrogen bond donor, including quaternary ammonium salts in combination with hydrogen bond acceptors like urea, glycerol, amides, or amines [48, 49]. Eutectic basically refers to a specific ratio between hydrogen bond donors and acceptors that results in a mixture that yields a melting point lower than the two individual components [32]. Generally, DESs have relatively low vapor pressures, good thermal stabilities, and, for electrochemistry, wide windows of electrochemical stability [50]. There are five distinct categories of compounds that can be broadly split into. Type I compounds include such complexing agents as $ALCl_3$ with choline chloride, which has melting points far above water, seeing rare use with biomass. More favorable, on the other hand, for Type II DES processing of lignocellulosic biomass would be the hydrated metal halides and quaternary ammonium salts, which present better stability against oxygen and water.

Hydrogen bond donors like alcohols, amines, and amides, which can easily be modified using a large array of donors, have proved feasible in synthesizing Type III DESs [51]. This is because these components are easy to synthesize and biodegradable. Type IV DESs, on the other hand, are formed by mixing transition metals with hydrogen bond donors such as urea; however, the ionization of metal salts in nonaqueous solutions may be challenging. Recent studies have demonstrated that DESs Type V are hydrophobic and nonionic, an interesting area that should be further researched due to their high potential for lignin fractionation [52]. The DES combinations listed in Table 1 were selected based on their frequent usage and proven effectiveness in biomass pretreatment studies [53]. These combinations represent a spectrum of hydrogen bond donor (HBD) and acceptor (HBA) with varying

Table 1 Different types of DESs based on the specific combinations of salts and hydrogen bond donors used

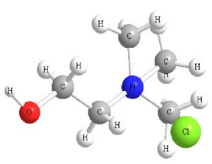
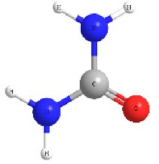
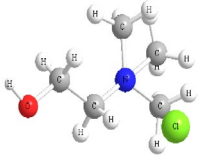
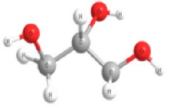
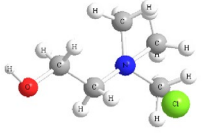
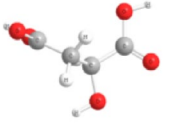
DES Types	Combination	Applications	Structure	References
Choline chloride/urea	Choline chloride (ammonium salt) with urea (a hydrogen bond donor)	Biomass pretreatment and extraction processes	 **Choline-Chloride**  **Urea**	[54, 55]
Choline chloride/ glycerol	Choline chloride and glycerol	Solvent extraction, catalysis, and electrochemical processes	 **Choline-Chloride**  **Glycerol**	[56, 57]
Choline chloride/ malic acid	Choline chloride with malic acid	Ability to dissolve lignocellulosic biomass, making it useful for pre-treatment in biofuel production	 **Choline-Chloride**  **Malic acid**	[58]

Table 1 (continued)

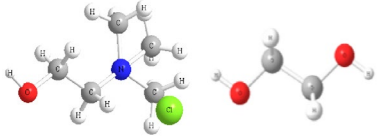
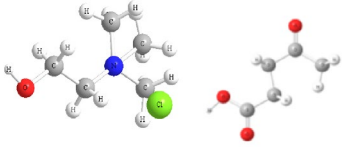
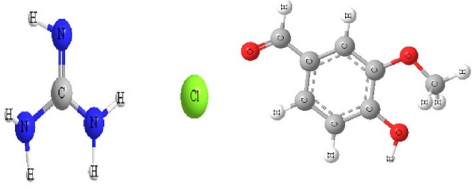
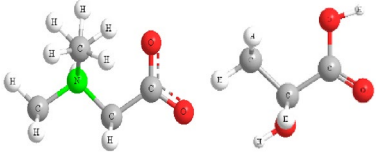
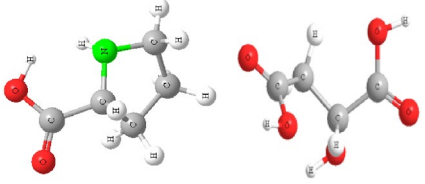
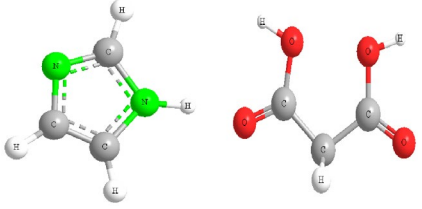
DES Types	Combination	Applications	Structure	References
Choline chloride/ethylene glycol	Ethylene glycol with choline chloride	CO_2 capture, lubrication, and electrochemical processes	 **Choline-Chloride** **Ethylene glycol**	[59, 60]
Choline chloride/levulinic acid	Levulinic acid (a hydrogen bond donor) and choline chloride	Applications such as biomass conversion and enzyme immobilization	 **Choline-Chloride** **Levulinic acid**	[61]
Guanidine hydrochloride/vanillin	Guanidine hydrochloride with vanillin	Used in the selective extraction of lignin and aromatic compounds	 **Guanidine Hydrochloride** **Vanillin**	[62]

Table 1 (continued)

DES Types	Combination	Applications	Structure	References
Betaine/lactic acid	Betaine (a hydrogen bond acceptor) and lactic acid	Improved enzymatic hydrolysis	 **Betaine** **Lactic Acid**	[63]
Proline/malic acid	Proline (a hydrogen bond acceptor) and malic acid	Efficient in lignin removal and biomass delignification	 **Proline** **Malic Acid**	[64]
Imidazole/malonic acid	Imidazole with malonic acid	Efficient lignin removal from corncob and improved sugar yields	 **Imidazole** **Malonic Acid**	[65]

acid–base characteristics, polarity, and biodegradability. These systems are extensively reported for their superior ability to selectively dissolve lignin, for enzymatic hydrolysis, and maintenance of carbohydrate fraction. Moreover, these DESs are very low-cost, easy to prepare, and exhibit a recyclable nature, making them attractive candidates for sustainable applications in biorefinery. These solvents exhibit chemical properties such as low volatility, high chemical and thermal stability, biodegradability, easy storage and transportation, and nonflammability [47].

Research has been done in terms of assessing the impact of deep eutectic solvent pretreatment on cellulose enzymatic hydrolysis and subsequent sugar yields. The results showed that the yields of glucose and xylose were higher in the choline chloride/urea pretreatment than in other pretreatments, such as dilute acid and ionic liquids. Choline chloride/urea DES has been studied widely regarding its effectiveness in pretreating lignocellulosic biomass [66, 67]. This removes lignin and reduces cellulose crystallinity, thus improving cellulolytic enzymes' accessibility to the biomass. Lignin removal and cellulose crystallinity reduction have been proven to be effective in DES [68–70]. Nonetheless, DES pretreatment remains a hot research area where much work is still necessary in order to optimize the conditions for the process and underlying mechanisms and scale it up to industrial applications.

Physicochemical and functional attributes of DESs

The physicochemical and functional properties of DESs have been applied in a wide range of applications. For example, in DES, it plays different roles between HBA and HBD as suggested [71]. Traditionally, the HBA will be a Lewis acid, while the HBD is a Lewis base [72]. This creates a specific physicochemical eutectic mixture when two constituents in a defined molar ratio are mixed together. DES can be manipulated in the ratios and concentration of HBD and HBA to work in most applications [73]. Intermolecular interactions basically determine the physicochemical properties of the hydrogen bond donor and acceptor. From the available scientific reports, it is concluded that hydrogen bonding interaction is the fundamental interaction involving the constituents of DESs. Ionic interaction is crucial for inorganic or organic salt-containing DESs, which come in Types I to IV, but they are most important for Type V DES. These molecular interactions, especially involving hydrogen bonds, significantly impact the viscosity of DESs, an important physicochemical property that dictates mass transfer and fluidity in a solvent for a particular chemical reaction. Low-viscosity solvents are often in demand for these reasons.

The vast majority of DESs have high viscosity values; this is one of their primary drawbacks [74]. Generally, DESs are denser than water, and the density values depend on temperature. The molar ratio and the structural features of the DES constituents exert a significant influence on density [71]. Figure 1e represents the temperature dependence of the density and viscosity for some typical DESs. In addition, DESs have very low vapor pressure. They are relatively nonvolatile and more environmentally friendly as compared to traditional organic solvents. In addition, DESs exhibit excellent thermal stability that allows their use over a wide range of temperature conditions without significant degradation [75, 76].

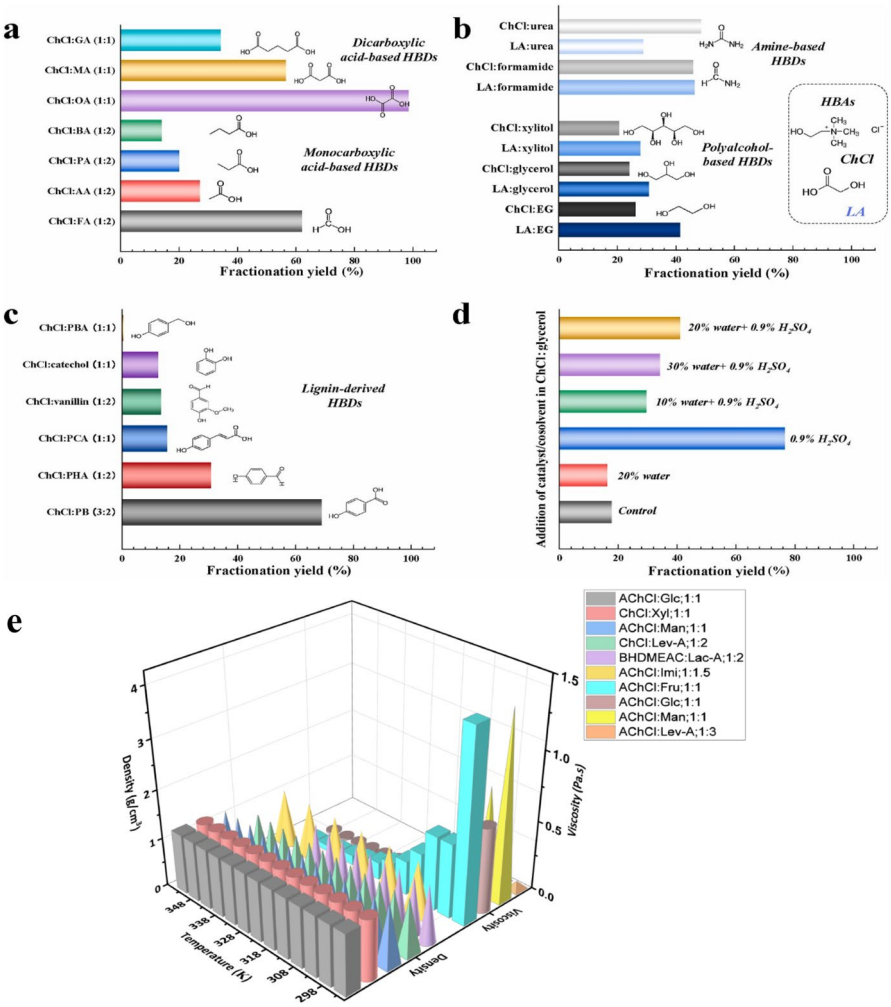


Fig. 1 The effect of DESs on lignin fractionation yield: **a** number of function groups and alkyl chain length in HBD, **b** type of functional group in HBA, **c** lignin-derived HBD, and **d** addition of catalyst/cosolvent in ChCl/glycerol (1:2) reprinted from [51] with permission from Elsevier and **e** effect of temperature on the density and the viscosity of some commonly used DESs, data derived from DES solution [79]

DESs have an enormous range of functional properties, which makes them versatile solvents. They can dissolve a wide range of polar and nonpolar compounds, which makes them amenable to a wide variety of applications, such as extractions and separations [77]. Moreover, DESs can form inclusion complexes with guest molecules, in which the stability and solubility of some compounds could be improved [78]. Hydrogen bonding interaction between a hydrogen bond acceptor, such as the chloride ion of choline chloride, and a hydrogen bond donor, which includes the hydroxyl group of ethylene glycol, is the key mechanism in DES

formation. Chemical reaction and compositions for DESs are usually diversified and tailored to various combinations between HBA and HBD that may be investigated. The choice of specific HBA and HBD may significantly influence the properties of the resulting biomass fraction, as shown in Fig. 1a–d.

The Kamlet–Taft polarity parameters, namely, α, β, and Π^*, have drawn a great deal of emphasis among various solvent properties of DESs [80]. A few studies have investigated the correlation between the K–T parameters and efficiency in the disintegration of cellulose, hemicellulose, and lignin. In these studies, it indicates that α, β, or β–α are in a positively correlated relationship with pretreatment efficiency [81]. However, there are limitations in using a straightforward linear correlation to assess K–T characteristics and the efficacy of DES-based treatments. To validate the previously proposed notion, earlier work has sought to develop a machine learning model to examine the relationship between K–T parameters and lignin degradation. In recent years, machine learning has emerged as a vital tool in the field of high-value lignocellulosic biomass utilization [82]. By incorporating the findings from molecular simulations as descriptors in the machine learning model, the solvent–biomass interaction can be more thoroughly investigated. This approach offers an opportunity to explore the unique intrinsic relationship between the macroscopic variables observed in experiments and the microscopic variables estimated through quantum chemistry [28].

This implies that the solubility of lignin in DES could be well predicted by including solvent properties defined by the Kamlet–Taft (K–T) model. Usually, one of the crucial properties that control the lignin dissolution process is the high HBA capacity of the solvent. Liu et al. [83] performed a detailed study in order to investigate the role of K–T parameters of DES on the solubility of lignin (Fig. 2). Their results indicated that the α and β values were more closely related to lignin solubility than the π^* value. More precisely, for Type III DESs, the ability of a DES to dissolve lignin was positively related to the difference between α and β values. For Type V DESs, however, the relationship between lignin solubility and K–T parameters is more complex. Given that it is easy to modify and measure the K–T parameters of DESs, optimization of DESs for efficient extraction of lignin from lignocellulosic biomass carries great potential for improvement in pretreatment processes of biomass.

Investigation of DES-based approaches for biomass fractionation

In DES-based biomass fractionation processes, a solvent or reaction medium is used to biomass for the separation of its components [84]. These processes aim to break down the biomass into its components (cellulose, hemicellulose, lignin, and other valuable compounds. In comparison with other solvents, a major advantage of DESs is their capability to fractionate lignocellulosic biomass, dissolving (or selectively extracting) specific components under mild conditions [85]. The first step of the experimental process is to choose a suitable deep eutectic solvent for lignocellulose biomass pretreatment [86].

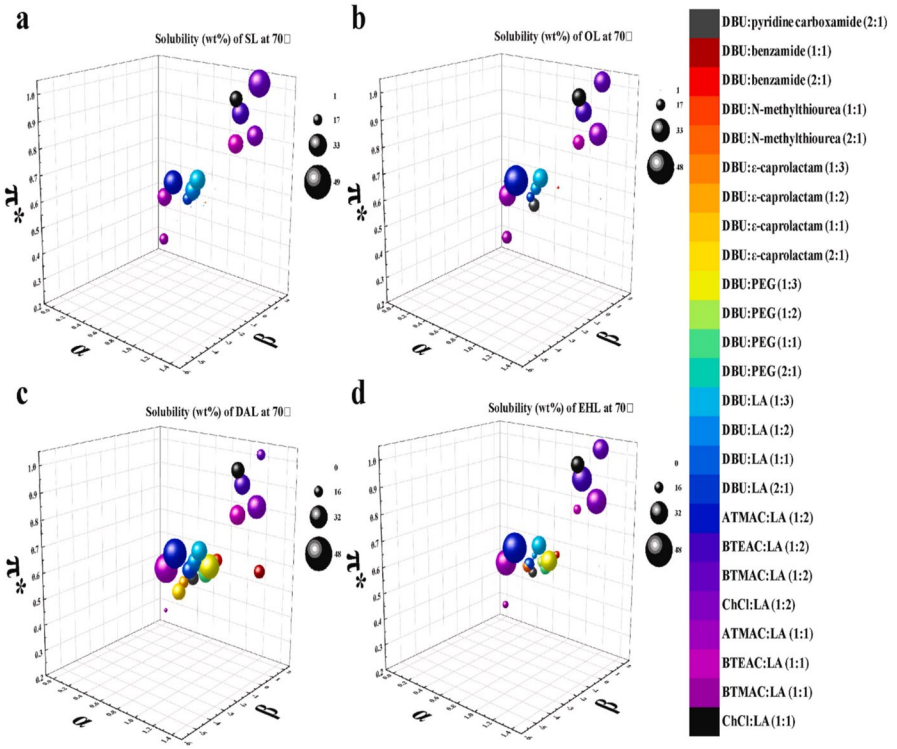


Fig. 2 The relationship between four types of lignin solubility and DESs' K–T solvatochromic parameters (lignin solubility is represented by ball size) reprinted from [51] with permission from Elsevier

Lignin extraction from biomass using DESs

Among green solvents, DESs have also been highly effective in biomass delignification due to their modifiable acidity as well as strong hydrogen bonding interactions [87]. Several studies have demonstrated that DES pretreatment has the ability to decrease cellulose crystallinity significantly but preferentially cleaves lignin-carbohydrate bonds [88]. Acidic DESs particularly facilitate lignin extraction through protonation of benzylic carbon atoms, encouraging ester and ether bond cleavage with subsequent depolymerization and potential re-condensation of the depolymerized lignin fragments [89]. Delignification performance is enhanced with water or catalytic additives that reduce viscosity and enhance mass transfer. Here, anti-solvents such as ethanol or acetone facilitate lignin recovery, whereas co-extraction and material corrosion remain problems of practical interest. In spite of the heterogeneous and recalcitrant nature of lignin, the selection of suitable DES types and operating conditions continues to be critical for efficient and selective delignification [90].

Choline chloride (ChCl)-based DESs have been effective at removing lignin because chloride ions (Cl^-) establish hydrogen bonds with polysaccharides,

disrupting the lignin-carbohydrate complex [90]. Lignin separation efficiency is determined by the position of hydroxyl groups in the biomass and the content of hydrogen bond donors (HBD), with excessive hydroxyl groups causing a rise in DES viscosity and extraction resistance [51]. HBA structural characteristics, such as functional groups and alkyl chain length, influence the removal of lignin, too. More effective removal is brought about by shorter alkyl chains of HBA due to the lower steric hindrance [91]. Betaine/lactic acid and ChCl-based DESs have been effectively exhibited for efficient lignin removal by researchers [92].

Lignin, the primary aromatic polymer in lignocellulosic biomass, accounts for over 30% of the world's organic carbon [93]. Traditionally, it has been burned for heat and electricity generation in second-generation biorefineries [94]. However, lignin has significant potential for valorization, with applications in the production of polymers, carbon fibers, and adhesives, depending on its quality. It can also be utilized to obtain monomeric phenolic compounds, such as 2-methoxy-4-propyl phenol, for use in diesel engines [95]. Compared to traditional harsh chemical methods, DES-based separation offers a relatively mild and environmentally friendly approach for lignin extraction. The solubility of lignin in DESs allows for selective extraction, leaving other biomass components like cellulose and hemicellulose behind [96]. Figure 3 schematically illustrates the process of dissolving lignin in a DES, creating an environment suitable for its separation.

Hemicellulose extraction from biomass using DESs

Hemicellulose, in comparison to cellulose, exhibits limited solubility in DESs. As a highly sensitive biomass component, hemicellulose can be easily hydrolyzed or degraded in acidic or alkaline environments. The strong covalent and hydrogen bonds between lignin and carbohydrates, through benzyl ether, benzyl ester, and

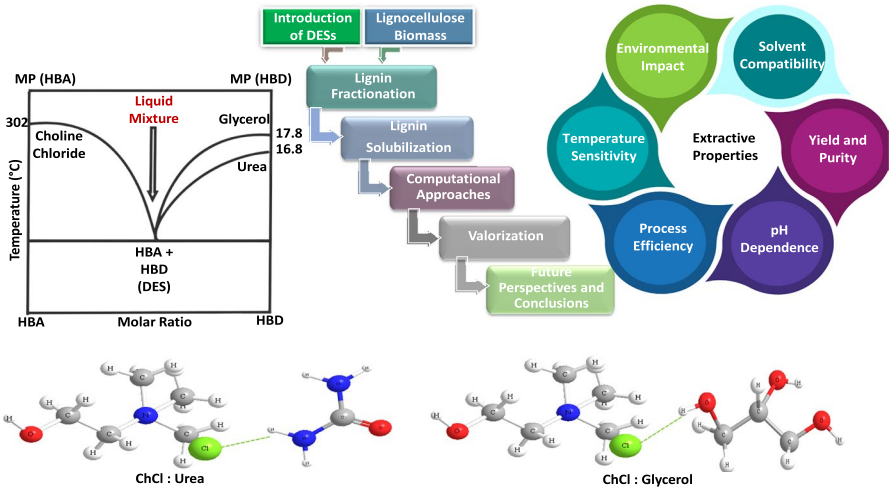


Fig. 3 The schematic of dissolving lignin in a DES suitable for the separation of lignin adapted from [97] with permission from John Wiley and Sons

benzo-side functional groups, contribute to the resistance of biomass [98]. As a result, the main target of the majority of the pretreatment process, in this case, is to disturb the bonds of the lignin and carbohydrate complex in order to make the extraction of hemicellulose easier. [99]. Hence, the efficiency in the removal of hemicellulose relies on the composition of DES used in lignocellulosic biomass form and pretreatment conditions. DESs extract hemicelluloses readily from biomass. It could extract hemicellulose largely under neutral circumstances and cause little degradation or deacetylation in the depolymerization of lignocellulosic biomass. The acidic DESs, nevertheless, are more competent at breaking the lignin and hemicellulose ether linkages as well as ester bonds, thus facilitating the dissolution of hemicelluloses, though it might simultaneously result in depolymerization. DESs could be fine-tuned to target hemicelluloses over lignin, which can improve the solubilization as well as the extraction capacity of some pretreatments. An example of this can be observed in several cases where DESs such as choline lysine/urea and one based on choline chloride/organic acid are effective at extracting cellulose while achieving reasonable removal of hemicelluloses even at mild temperature settings, but the lignin is poorly or not extracted from a mirrored range biomass source. The four hemicellulose extraction methods demonstrate varying efficiencies and

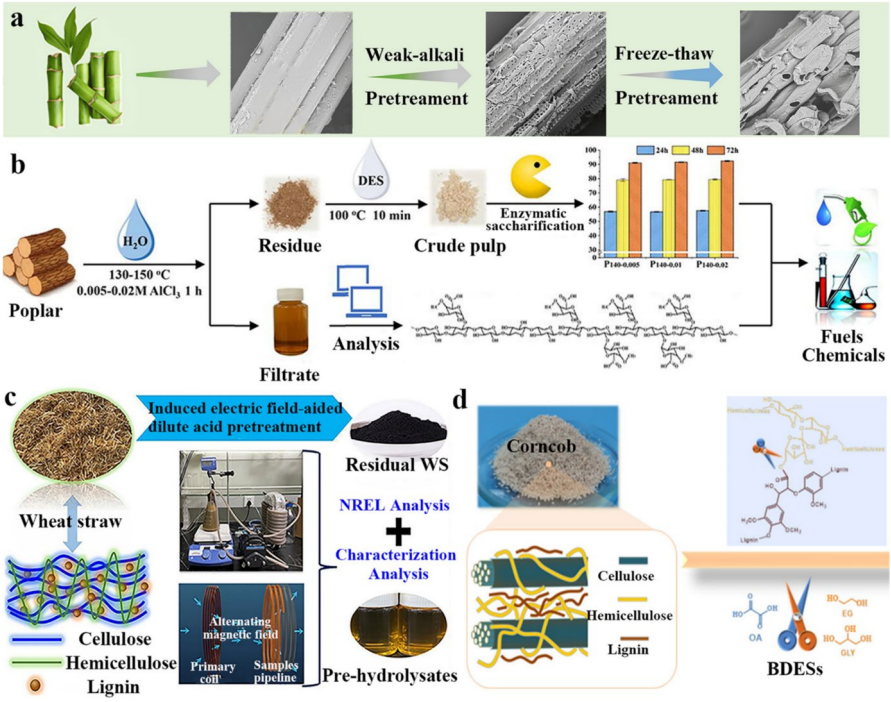


Fig. 4 Hemicellulose extraction with various methods. **a** Extraction of hemicellulose with freeze–thaw-assisted alkali process. **b** Hydrothermal combined with DES for the extraction of hemicellulose. **c** Induced electric field-assisted dilute acid pretreatment. **d** Deep eutectic solvents pretreatment reprinted from [100] with permission from John Wiley and Sons

are described by Cheng et al. [100] as shown in Fig. 4. Freeze–thaw-assisted alkali yields about 60–70% through strengthening the cell wall and weakening. Hydrothermal + DES yields 75–80% recovery through heat loosening with subsequent selective DES extraction. Electric field-assisted dilute acid yields 65–70%, with a boost from enhanced ion transport. Direct DES pretreatment gives the best yield at 70–85% under gentle, green conditions. This suggests that DES-based protocols, especially when used in combination with hydrothermal treatment, provide the highest yield and selectivity and are therefore ideal for sustainable and structure-sensitive materials like hydrogels.

Finally, the heating technology used would also be a critical issue based on previous studies, where it was reported that microwave-assisted heating, along with specific DES formulations, was capable of higher hemicellulose removal rates compared to lignin. The obtained results on the interactions enable comprehension of hemicellulose extractability and yield, resulting in insights for future designs of high-efficiency biomass valorization pathways. Several steps are involved in the extraction of hemicellulose from biomass. Biomass enters an extraction step after appropriate pretreatment techniques, through the use of water, organic solvents, or DESs as selective solvents for hemicellulose recovery [101, 102]. Filtration techniques, such as centrifugation or membrane filtration, separate the extracted hemicellulose [103]. This hemicellulose can then be further purified by chromatography, precipitation, or other methods to further purify it. Lastly, the hemicellulose is dried into a solid product that can either be stored or, if required, further processed for value-added applications. Each step of this process must be optimized in order to effectively extract hemicellulose and achieve the highest value from the biomass for a wide variety of applications.

Cellulose fractionation from biomass using DESs

The strong cohesive forces and the high degree of intermolecular hydrogen bonding in cellulose make its dissociation and solubilization rather difficult in DESs [104]. However, by considering the distinct properties of DESs, especially with the choline chloride formulation, researchers have been pretty successful in stabilizing the DES/cellulose system through the formation of a very effective hydrogen bonding network [105]. This mechanism entails the disruption of existing hydrogen bonds in cellulose and the establishment of new hydrogen bonds between the cellulose and the DES components. Traditionally, acidic DESs have been widely employed for biomass fractionation purposes, focusing on the removal of lignin and hemicelluloses, as highlighted in previous comprehensive reviews [106]. DESs cause dissolution of cellulose through the disruption of intermolecular hydrogen bonds and the formation of new hydrogen bonds between cellulose hydroxyl (-OH) groups and DES molecules [91]. Weakening of inter- and intramolecular hydrogen bonds in cellulose is promoted in DESs by the HBA and is correlated with DES hydrogen bond basicity (β). This basicity plays an important role in determining the dissolution performance of cellulose in DES and ionic liquid (IL) pretreatment. Nevertheless, β is considerably altered through the addition of anions like Cl^-, HCOO, OAc^-, and

$(MeO)_2PO_2^-$. Among the range of DESs, choline chloride (ChCl)-based DESs are efficient HBA for dissolving cellulose and hemicellulose in lignocellulosic biomass [107].

Recent studies have investigated various deep eutectic solvent formulations, yielding promising results. As shown in Table 2, these studies have explored the biomass source, DES type used for pretreatment, processing conditions, and the ratios of extracted biomass components. For instance, in biomass fractionation, a combination of choline chloride and glycerol outperformed the use of urea and ethylene glycol, resulting in a cellulose-rich fraction with exceptional purity [54]. The preference for the ChCl/glycerol DES can be attributed to the lower acidity and polarity of glycerol, which enhances the solubilization of amorphous cellulose [68]. Moreover, the cellulose fractionation depends on the temperature. Based on some experiments, it has been reported that cellulose recovery, grading, and regeneration peaked at approximately 107 °C. In addition, many research studies prove that DESs are capable of separating amorphous cellulose from herbaceous biomass with hemicelluloses, sugars, and lignin content depending upon the composition used in DES and the particular conditions during its application [108].

Sustainable chemical production from biomass using DESs

Biomass is turning out to be a highly accepted feedstock for renewable and abundant sources of sustainable chemical production. Scientists are currently researching DESs to devise revolutionary means of extracting key elements of biomass and further transmuting them into high-value compounds. Such solvents have some potential advantages because they can dissolve and separate each component of biomass, like lignin, cellulose, and hemicellulose, in an easy and selective way.

Extraction of high-value compounds from biomass

DESs are versatile systems applied for the extraction of desirable compounds from biomass. Here, valuable compounds range from antioxidants and pigments up to bioactive pharmaceutical entities [120]. One of the more significant advantages of DES as an extraction medium is its ability to recover such chemicals before additional pretreatment schemes on the biomass occur [121]. This reduces the risk of contamination of major biomass components and prevents such chemicals from degrading or being lost under conditions usually encountered in downstream processing. In addition, DES-based extraction is particularly appropriate for the manufacture of "clean label" compounds with antioxidant, coloring, or functional functionalities. Such applications are of more interest to natural DESs because they are intrinsically derived and biocompatible in their formulation, hence directly applicable in food, pharmaceutical, or cosmetic applications [122, 123]. DESs also selectively recover target compounds such as phenolics, sugars, or lignin with reduced co-extraction of impurities. This decreases the number of downstream purification steps from conventional solvents that have a tendency to extract broader chemical profiles [124].

Table 2 Biomass DES type used for pretreatment, processing conditions, and the extracted components ratios of biomass

Feedstock	DES type	Molar ratio	T °C), t (min)	Lignin (%)	Hemicellulose (%)	Cellulose (%)	Reference
Poplar bleaching pulp	Amino sulfonic acid: glycerol	1:3	100 °C for 1–1.5 h	NR	NR	80.9–95.2	[109]
Mosso bamboo	ChCl/succinic acid	2:1	160 °C, 10 min	6.2	86.4	44.8	[110]
Mosso bamboo	ChCl/OA	2:1	160 °C, 10 min	42	58.6	48.2	[110]
Bleaching eucalyptus kraft pulp	ChCl/oxalic acid	1:1	80 °C, 2 h	NR	NR	82.3	[111]
Maritime pine sawdust	ChCl/LA	1:5	150 °C, 2 h	81.3	NR	NR	[94]
Maritime pine sawdust	ChCl/LA	1:2	150 °C, 2 h	76.7	NR	NR	[94]
Coconut coir	ChCl/LA	1:4	150 °C, 20 min	82	NR	NR	[112]
Rice straw	ChCl/acetic acid	1:3.6	126 °C, 2.5 h	83.1	82.1	NR	[113]
Raw poplar sawdust	$ZnCl_2$/LA	1:10	140 °C, 3 h	97.5	NR	NR	[114]
Bamboo residues	ChCl/oxalic acid	1:1	130 °C, 6 h	21.2	91.1	9.67	[115]
Poplar wood	Cholinium lysine/urea/5 wt% water	1:2	90 °C,6 h	NR	35.3	62.13	[115]
Poplar wood	K/1, 2-PG	1:2	130 °C, 7h	89.2%	71.6%	97.5%	[115]
Miscanthus	ChCl/EG	1:2	110 °C, 12 h	16.7	25	53.1	[116]
Oil palm fronds	ChCl/$Cacl_2.6H_2O$	1:2	125 °C, 4.5 h	44.53	NR	NR	[117]
Mosso bamboo	ChCl/Fecl3/GLY	1:2	120 °C, for 2 h	62.77	76.6	NR	[118]
Miscanthus	ChCl/acetic acid	1:2	150 °C, 3h	9.5%	NR	NR	[119]

Various DES combinations, such as ChCl and betaine with organic acids, urea, or carbohydrates, as well as carbohydrate mixtures, have been utilized to recover high-value molecules. Ternary DES formulations, like ChCl/levulinic acid/N-methyl urea, have also demonstrated effective outcomes, particularly when the target extraction involves a diverse collection of phenolic compounds with varying polarity [125]. The polarity of the DES influences the solubility of phenolic compounds, with moderately polar DES successfully dissolving weak and medium polar phenolic compounds through the "like dissolves like" principle. The water content in DESs can easily be adjusted to vary the DES polarity index, thus increasing extraction efficiency, but the hydrogen bonding interaction with target chemicals at such water content reduces overall yield extraction [123, 126, 127]. In the efficient extraction of polymeric phenolic compounds like tannins, promising results have been shown using quaternary eutectic solvents containing ChCl, glycerol, ethanol, and water. The integration of DESs with microwave irradiation heating further optimizes the DES extraction process by reducing extraction time, increasing yield, and improving selectivity [128].

DES was demonstrated to be a viable technique for the extraction of less polar compounds, carotenoids, and triterpenoids, from all types of waste biomass. Such mixtures of choline chloride/tartaric acid with methanol, as cosolvents, ethyl acetate, and ethyl lactate or hydrophobic DES from natural terpenes can be effective in the process of extraction [129]. The application of DESs for the sustainable extraction of high-value chemicals from biomass presents an intriguing path for the synthesis of clean and beneficial substances [130]. This application of DESs has interesting prospects for the development of clean and useful derivatives from biomass. Thus, DES formulations are versatile in their formulation and broadened extraction capabilities to be quite an important tool in achieving sustainable chemical synthesis from biomass [131].

Reducing sugars from biomass

Deep eutectic solvent extraction is a promising system that may be used to extract reducing sugars from biomass; hence, fractionation from lignin and hemicelluloses is adopted [116]. In this process, the cellulose-rich solid fraction is separated from the liquid fraction containing the solubilized components, such as lignin, hemicelluloses, and their degradation products. Following this separation, enzymatic hydrolysis is employed to convert the cellulose into glucose [132]. Zhang et al. [133] studied the effects of pretreatment time Fig. 5a, temperature Fig. 5b, $FeCl_3$ loading Fig. 5c, and ChCl/NE dosage Fig. 5d on the corn stover (CS) pretreatment and determined the optimal conditions to maximize the reducing sugar yield. Enzymatic hydrolysis of the DES-pretreated CS showed a significant increase in saccharification efficiency as compared to that of untreated CS, as shown in Fig. 5e. The addition of 30% water into the DES system improved the pretreatment of biomass through the reduction of solvent viscosity, thus allowing better mass and heat transfer. Among the tested DESs, the performance of the ChCl/NE/30% water system was the highest, giving a yield of 80.3% for the

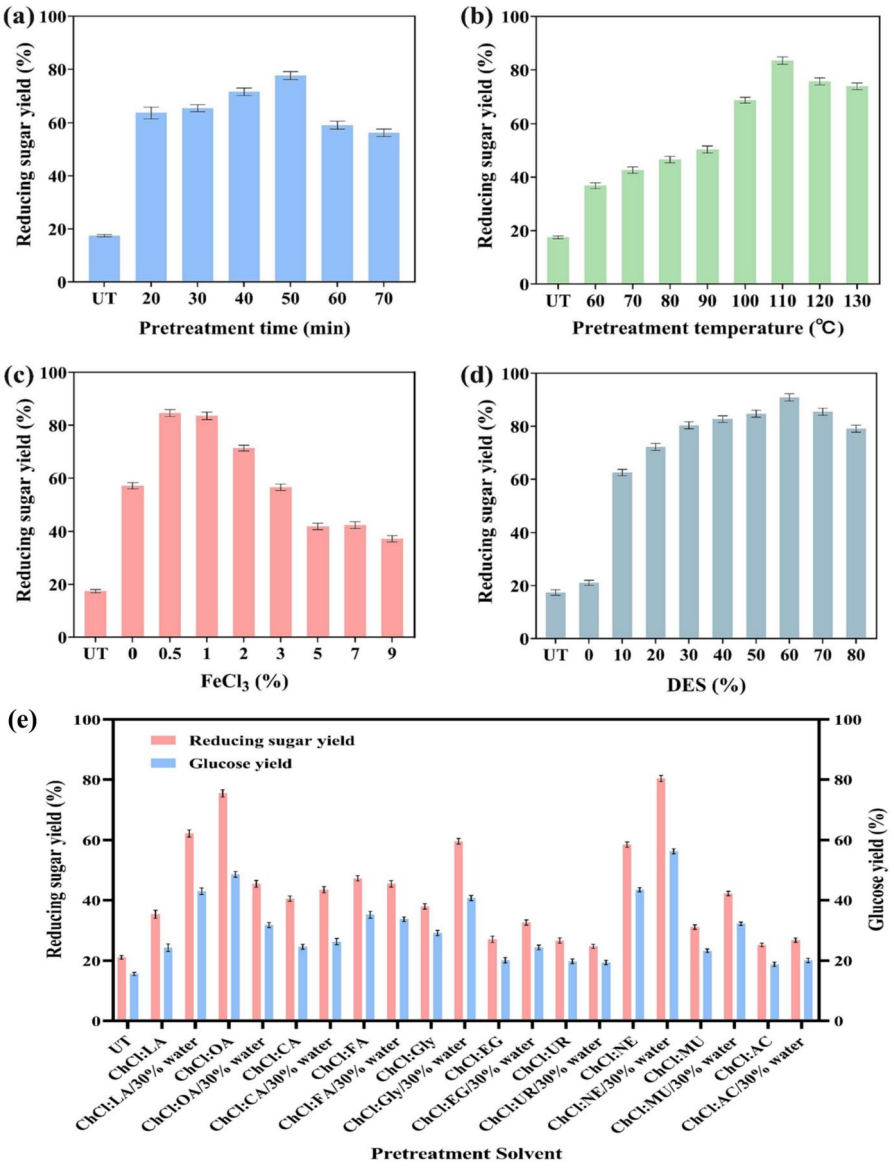


Fig. 5 The effect of various pretreatments **a** time, **b** temperature, **c** $FeCl_3$ loading, **d** DES content, and **e** comparison of various DESs with ChCL as the HBA and different HBD on the corn stover pretreatment reprinted from [133] with permission from Elsevier

reducing sugar and 56.2% for glucose at 28.1 g/L. Acidic DESs were also performing well, such as ChCl/OA/30% water, which yielded 75.5% reducing sugars and 31.8% glucose. Neutral DESs like ChCl/Gly yielded 59.6% reducing sugars and 40.7% glucose. These results indicate that the acidity of the HBDs and the

optimal processing parameters are key factors in the enhancement of pretreatment efficiency, with acidic and alkaline DESs being more effective than neutral ones.

DESs with the acid-based hydrogen bond donor have also been widely employed for the lignin and hemicellulose removal process that results in improved cellulose saccharification and further glucose yields [134]. Other organic acids, even shorter chains, such as formic acid and acetic acid, have also been added to DES systems to enhance the conversion of cellulose into glucose. The yield has reached a maximum of 92%. It turned out to be an essential step of DES pretreatment in minimizing the surface lignin strength and condensable strength for easy enzyme availability to the cellulose [113]. So far, the use of conventional heating was mainstream during the DES process, while microwave-assisted heating depicted good hopes in order to achieve superior enzymatic efficiency with less energy consumption. For instance, the combination of microwave-assisted heating and a ternary DES consisting of ChCl/ethylene glycol: AlCl3 resulted in enzymatic saccharification yields up to 95.4%. The use of DES-based ultrasound-assisted pretreatment has also been applied for enhancing both delignification and subsequent enzymatic hydrolysis yields [112]. To counteract the increased viscosity of DESs, which can limit the cavitation-disrupting impact associated with ultrasonic irradiation, the addition of modifiers such as water or alkaline aqueous solutions has been shown to advance the sugar synthesis process [135].

Sustainable biofuel production from biomass

DESs are a good potential solvent for many chemical transformations in the biorefinery. They offer an appropriate environment for the conversion of biomass-derived mixtures into valuable compounds and biofuels [136]. DESs have been used for the synthesis of platform chemicals such as levulinic acid and furfural, which can further be converted to a variety of bio-based products. Several routes have been explored in the literature toward alcohol biofuels, mainly on glucose feedstocks. One example is the processing of miscanthus biomass with a mixture of dimethyl sulfoxide and choline chloride/glycerol DES to generate a cellulose-rich solid. This material is subsequently hydrolyzed and fermented, resulting in a high ethanol yield of approximately 72% [116]. Another crucial application is biodiesel manufacturing, where microalgae biomass serves as an appealing lipid source that does not compete with the food supply chain [137, 138]. There were conclusions that demonstrated an application of acidic-based DES in methanol media catalyzed by sulfuric acid that could improve biomass fractionation, lipid recovery, and concentration of the fatty acid methyl ester (FAME). However, the direct conversion of FAME from lipids wet biomass by the application of DES, reduces the need for hydrolysis and makes the general process much more efficient [139, 140]. The biodiesel production flow diagram from the biomass is shown in Fig. 6.

Several studies have reported enhanced bioethanol yield from lignocellulosic biomass sugar hydrolysates that have been pretreated using various DESs. For example, Vaid et al. [142] reported bioethanol production at 173.61 mg/g from calcium hydroxide-supported ChCl/glycerol pretreated Saccharum spontaneum biomass in

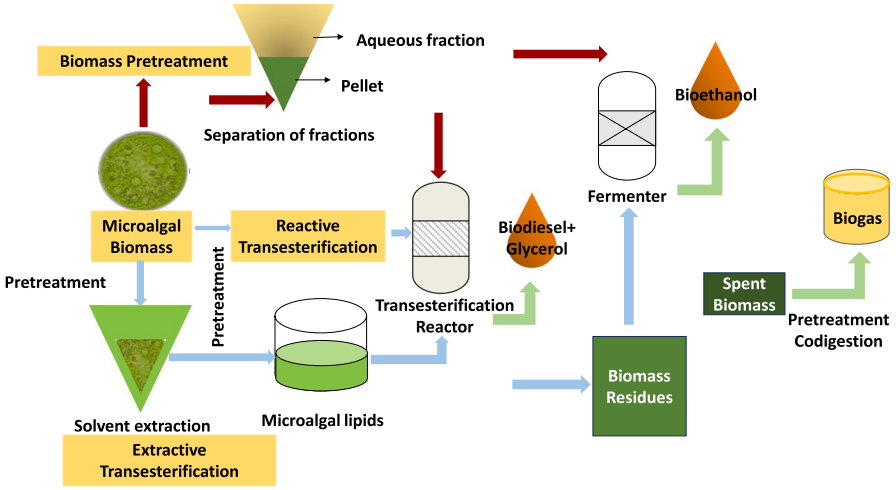


Fig. 6 The Integrated Bioprocess for Biofuel Production from Microalgal Biomass includes transesterification, fermentation, lipid extraction, and biomass pretreatment reprinted from [141] with permission from Elsevier.

an integrated one-pot bioprocess. Similarly, Xu et al. [143] pretreated corn stover with ChCl/glycerol and achieved 149 g/kg biomass ethanol yield at 77.5% theoretical yield from the initial glucose content. In addition, Okuofu et al. [144] stated that Bambara groundnut haulm pretreated with ChCl/lactic acid resulted in 60.7% delignification, 94.8% sugar recovery, and a maximum ethanol concentration of 11.57 g/L with a yield of 0.38 g/g sugar through fermentation using Saccharomyces cerevisiae. DESs have also been employed for the enzymatic transesterification of vegetable oils for the purpose of producing biodiesel. He et al. [145] documented the production of biodiesel from lipase from Yarrowia lipolytica with a 91.8% yield under optimized conditions. Yield of biodiesel was enhanced to 98.5% using ChCl/glucose DES. Zhao et al. [146] in another article obtained high soybean oil triglyceride conversion (88%) to biodiesel using ChCl/glycerol under optimized conditions.

DES-based biomass fractionation systems have also been applied in the production of gaseous biofuels, which include biogas and biohydrogen [147]. The cascading fractionation, including hydrothermal and DES pretreatments, has effectively removed hemicelluloses and lignin, hence increasing methane generation during anaerobic digestion [148]. Besides, different DES formulations that include ethanolamine have proven their ability in delignification and hemicellulose removal, thereby making it easy to generate biofuel [137]. For instance, the DES of ChCl/ethanolamine pretreated corncob was used for the enzymatic hydrolysis and photofermentation of sugars that produced 151 mL/g biohydrogen. Furthermore, DESs can be an enzyme-stabilizing medium with improved catalytic activity and stability. Improved enzymatic conversion may be achieved in biorefineries. The use of DESs in the process of producing enzymes for biofuel and bioplastics has resulted in observations regarding increased process efficiency, as well as cost-saving [149].

These studies together conclude the potential DESs bear toward sustainable biomass-based fuel production. [140].

The main barriers to the commercialization of LCB-based biorefineries are the sustainable supply of biomass and variability in feedstock composition arising from geographic and seasonal factors. Biomass yield and biofuel production are very sensitive to feedstock type, and a homogeneous supply is important. Investigations have shown that optimal feedstocks vary by location, such as corn stover in Egypt or sugarcane bagasse in Indonesia [150]. The fragmented and uncoordinated supply chain of biomass at present is not able to sustain large-scale operation due to its infrastructure and transport-related issues [151, 152]. Some other solutions may include feedstock blending, densification, and biomass selection [153]. Storage also faces other problems than those described above, which impact yield and stability, such as biomass perishability and moisture content. Biomass with more than 20% moisture is unstable, while dried biomass below 10% is desired [154]. Transportation costs are also relevant to economic viability, especially for nonlocally sourced biomass, where pelletizing biomass can be a potential solution, even if additional processing is required [155]. Lignin content is another major barrier to the utilization of LCB. The removal of lignin during pretreatment is a costly and cumbersome process, consuming approximately 30% of the total production cost [156]. Efficient valorization or reuse of lignin can reduce the cost of production and enhance overall output in a biorefinery. Different pretreatment methods such as chemical, physical, biological, and physicochemical are being used to improve biomass conversion; delignification and lignin valorization are the prime focus areas for biorefinery sustainability improvement [157].

Recycling and sustainability of deep eutectic solvents

One of the prime advantages of DESs involves their recyclability, thus rendering them very attractive when compared with other solvents [105]. Recovery and recycling of DESs following biomass pretreatment, fractionation, and conversion to added-value products are of vital importance for their biomass matrix-disruptive power [134]. The first step in recovering DESs from biomass pretreated with DESs involves the removal of compounds extracted from biomass, which include lignin, hemicelluloses, and extractives [158]. Macromolecular fractions can usually be recovered by the addition of an anti-solvent such as water, ethanol, or acetone that changes the polarity of the media and provokes the precipitation of the solubilized components [159]. The most applied recovery strategy of DES involves the evaporation or distillation of the anti-solvent, which removes the precipitated compounds, allowing recovery of DES free of macromolecular or insoluble compounds [160, 161]. Nevertheless, this method does not fully eliminate soluble impurities, degradation products, and residual extractives but may instead build up through successive reuses and impact DES performance [162].

Kim et al. [163] performed a compositional analysis that showed biomass pretreated with recycled ChCl/LA (Fig. 7a) had an increased cellulose content from R0 to R1 and then gradually decreased in the subsequent cycles. For ChCl/Gly, the

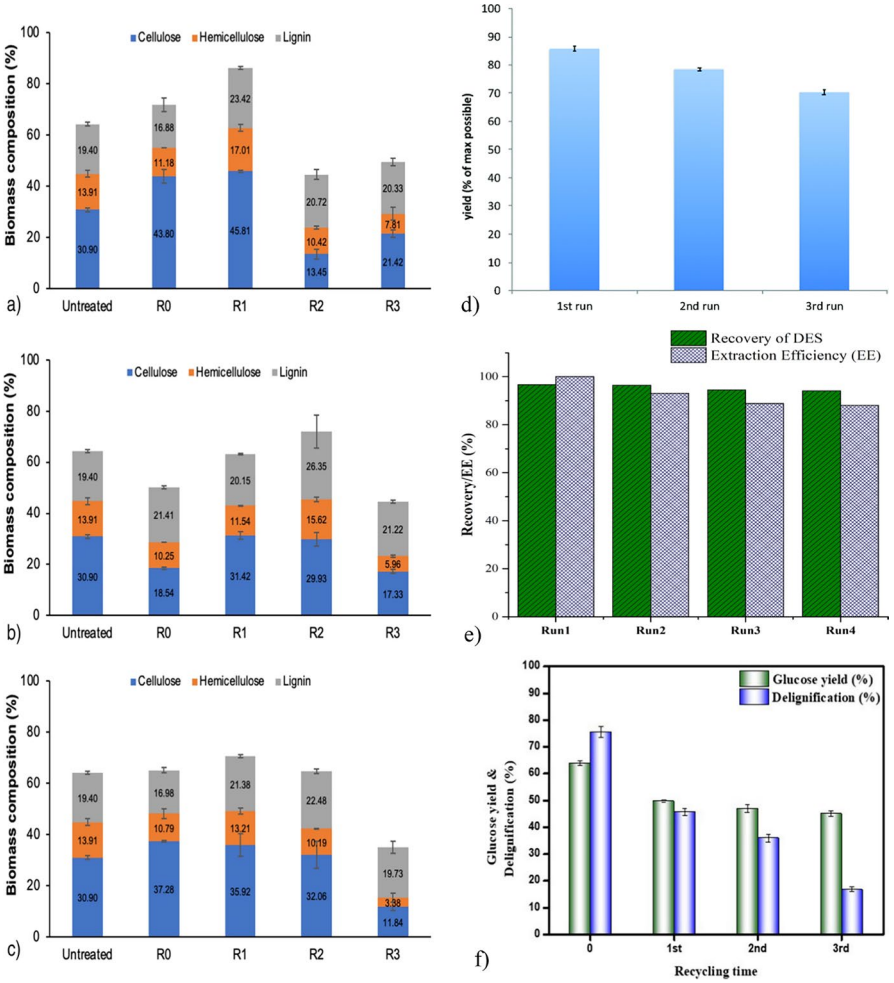


Fig. 7 Biomass fractionation in Napier grass pretreated with **a** recycled ChCl/LA, **b** recycled ChCl/Gly, **c** recycled ChCl/urea reprinted from [163] with permission from Springer Nature, **d** saccharification yield from the recycled ChCl/PCA pretreated switchgrass reprinted from [164] with permission from the Royal Society of Chemistry, **e** yield of the recovered DES and extraction efficiency after each cycle in chestnut shell waste reprinted from [165] with permission from ACS, and **f** the glucose recovery and delignification of bamboo under various ethanol/DES system pretreatment cycle times reprinted from [166] with permission from Elsevier.

cellulose content first decreased in R0, increased in R1, and then steadily decreased from R2 onwards (Fig. 7b). In the case of alkaline treatment (Fig. 7c), cellulose contents were higher than those in the untreated biomass until R2 but decreased in R3. Hemicellulose content was reduced within all DES pretreatments from R3 when compared with R0, but on the other hand, the content of lignin accumulated stepwise with recycling. This underlines the necessity of lignin and impurities removal

from recycled DES after every cycle. A number of methods have been reported for the recycling and recovery of DESs, such as evaporation, distillation, addition of an anti-solvent, membrane filtration, crystallization, solid/liquid extraction, liquid–liquid extraction, density separation, and supercritical fluid extraction [107].

The use of anti-solvents is a simple and effective way of recycling DESs after pretreatment of biomass. Anti-solvents, which are normally employed in high concentration, disrupt the hydrogen bonding between the DES components, leading to precipitation of the solubilized biomass constituents. Some of the common anti-solvents are water, acetone, and ethanol, which can easily be removed by evaporation so that the DES can be reused [167]. Mamilla et al. [84] used acetone for the separation of ChCl/oxalic acid DES from the reaction mixture after the pretreatment of beech wood sawdust. The DES was separated by drying water from the remaining mixture. Li et al. [168] concluded that ChCl/lactic acid DES showed good performance after being recycled five times with a 65% recovery rate at 90 °C when used for rice straw pretreatment. A 65% recovery rate after five recycling cycles indicates moderate recyclability compared to ILs [169]. The loss is due to the low thermal stability and boiling point of lactic acid [40]. Kumar et al. [170] also noted that the use of water as an anti-solvent for ChCl/lactic acid DES gave outstanding delignification when used in rice straw pretreatment. Membrane filtration methods have been investigated as another recovery option for DESs. For instance, Liang et al. [171] implemented a recovery system of ion-exchange membranes, ultrafiltration, and electrodialysis to segregate and recuperate ChCl/ethylene glycol DES following the hydrothermal pretreatment of eucalyptus globulus wood. Aside from this, ethanol was utilized as an anti-solvent in precipitating out the dissolved biomass compounds in DES.

Overcoming the challenge of impurities may thus become one of the keys to sustaining DES performance in successive cycles of pretreatment. Impurities can affect the entire process of biomass conversion [159]. A full assessment of DES recycling impacts must be addressed concerning its composition on the hydrogen bond acceptor/hydrogen bond donor ratio, total DES content, and low molecular weight soluble components [164]. Previous studies have reported a decrease in pretreatment efficacy when using recycled DESs, particularly acid-based DESs, which can experience losses of hydrogen bond acceptors and donors during regeneration, affecting the overall solvent acidity [172]. As shown in Fig. 7d and f, a modest decline in the glucose yield was detected with the growing number of DES recycles [164, 166]. Recycling alkaline DESs through acidification with CO_2 to precipitate extracted lignin has also demonstrated some reduction in delignification performance [173]. However, in certain cases, significant delignification can still be achieved after multiple recycling steps (Fig. 7e) [165]. While the composition and proportion of DES components can be maintained during recycling, a decrease in hemicellulose yield and molecular weight may occur [174]. The recyclability of DES offers both cost and environmental advantages. However, the volume of the anti-solvent that has to be used for washing and lignin precipitation can introduce economic challenges in the form of distillation costs [105]. Thus, great care is highly required when recovery and reutilization need to be balanced between sustainability and economic factors coming from recycling

DES [175]. Further research has to be done in order to understand and optimize the process of recycling and take into consideration changes in the composition of DES and how it can affect the properties.

Assessing the environmental and techno-economic impacts of des-based biorefineries

The sustainability of DES-based biorefineries requires techno-economic studies and environmental analyses to be conducted using life cycle assessment methods. Assessments give economic feasibility and environmental impacts of DES technologies, especially at low technology readiness levels during the early stages of development [176]. For DES-based biorefineries to be termed sustainable, they need to be technically scalable at an industrial scale and economically and environmentally favorable compared to competing processes [177]. LCA, among others, plays an important role since it aids in the quantification of the biorefinery system's environmental impacts. LCA addresses the entire life cycle, hence giving all the information needed for sustainable decision-making. Earlier research studies have been able to offer input on the challenges and suggestions concerning the application of LCA to biorefinery systems [178]. An LCA-based study on DES-based biorefineries would make a comparison in terms of environmental influences, such as different technologies, process conditions, or types of feedstocks used in biomass. The scope and limitations of the system would depend on the objectives set for the study. It is normally done by considering cradle-to-gate and cradle-to-grave assessments [179].

Economic viability is quite diverse for biomass pretreatment operations because they are dependent on the processing technique and economics. For example, a techno-economic assessment of bioethanol production from microwave-pretreated household waste reported production cost as 0.143 US$/L by Sondhi et al. [180], which made ethanol 8.32 times cheaper than commercial prices and hence rendered the operation highly viable. Similarly, Peng et al. 2021 [176] assessed the DESs-based pretreatment and reported the minimum ethanol selling price (MESP) as US$2128.1/ton for ChCl/LA, which is approximately USD 1.68 per liter, based on ethanol's density (0.789 kg/L). Sensitivity analysis showed that fluctuation in cellulose conversion rate and costs of acetone and LA had a sensitivity of 23.4%, 6.25%, and 5.31% toward MESP, respectively. Minimum selling price of BDO was significantly influenced by the pricing and recycling ratio of acetone and methyl isobutyl ketone in a study on the pretreatment of high solid loading DESs for furfural, lignin, and 2,3-butanediol production done by Zang et al. [181]. Future development should be done in a way of minimizes their consumption so that economic efficiency can be enhanced. Bussemaker et al. [182] studied sequential pretreatment methods using organic solvents and ultrasonic treatment. The model of Bio-sep Limited gave values for cellulose, hemicellulose, and lignin as GBP 2600, GBP 1200, and GBP 460 per ton with recoveries of 98%, 72%, and 94%. These high energy consumptions of 235 kWh for heat and 634 kWh for electricity suggested the possibilities of designing more efficient sonar reactors [183]. Economic feasibility in general depends on

Table 3 Summary of Economic and Environmental Assessments of DES-Based Biomass Pretreatment Processes

DES System	Feedstock	Assessment type	Indicators evaluated	Solvent recovery strategy	Key findings	Ref. No
ChCl/lactic acid	Agricultural waste	TEA	Minimum ethanol selling price (MESP), enzyme cost, and solvent cost	Recycled acetone/lactic acid	MESP: $2128.1/ton ethanol; cost sensitive to solvent and cellulose conversion	[176]
ChCl/DES blends	Mixed biomass	TEA	2,3-Butanediol (BDO) and furfural price	Acetone/methyl isobutyl ketone distillation	Solvent price and recovery rate significantly affect profitability	[177]
DES + IL + catalyst	Lignocellulose	TEA	BDO/lignin yield, energy use	Integrated solvent/catalyst loop	Solvent type and catalyst cost are critical for economic viability	[175]
ChCl/oxalic acid	Wheat straw	LCA	Emissions, energy intensity	Dilution/filtration	Identified DES synthesis and heating as environmental hotspots	[184]
ChCl/urea	Wheat straw (pellets)	Thermal + LCA	Greenhouse gas emissions, bioenergy yield	Low-temp evaporation	DES improves energy density, lowers emissions vs untreated straw	[185]
DES/IL system	Chestnut shell waste	TEA + Reusability	Yield per cycle, solvent retention	Recycle for 5 + cycles	High reusability, but lignin accumulation affects long-term efficacy	[165]
ChCl/oxalic acid	Beech wood	TEA	Cost of lignin recovery	Acetone anti-solvent + drying	Moderate-cost, efficient lignin extraction	[84]
ChCl/lactic acid	Corn stover	TEA + ML modeling	Cost, lignin solubility, and viscosity	Optimized via a predictive machine learning model	Identified low-cost DESs with high efficiency	[28]
ChCl/urea	Rice straw	TEA	Capital expenditure, operating expenditure, and sugar yield	Heat evaporation and reuse	25% cost reduction vs the alkali method	[163]
ChCl/levulinic acid	Agro-waste mix	TEA + Sensitivity analysis	Cost/ton, enzyme dosage, recovery yield	Anti-solvent recovery	Recyclability increased economic feasibility	[171]

Table 3 (continued)

DES System	Feedstock	Assessment type	Indicators evaluated	Solvent recovery strategy	Key findings	Ref. No
Various ChCl/acid DESs	Multiple lignocellulosic feedstocks	Review-based LCA/TEA metrics	LCA benchmarking, process energy	N/A	Identified scaling, viscosity, and solvent cost as key issues	[42]

solvent recovery, process optimization, and selection of an appropriate TEA model to obtain high biomass utilization and product value. A summary of economic and environmental assessments of des-based biomass pretreatment processes is given in Table 3.

Techno-economic assessments as well as environmental assessments via LCA methodologies are also important considerations in assessing sustainability in the case of DES-based biorefineries. The overall information concerning the economic feasibility, technological scalability, and environmental aspects of DESs associated with biorefinery is given through such reviews [186]. Figure 8 highlights the LCA process using a pretreatment method: DES-based biomass. Generally, it is reported to have major importance in various investigations in terms of selections for boundaries in system lifecycles for deep eutectic solvent-based biorefineries. The most commonly followed is the cradle-to-gate boundary, starting with raw material acquisition to the synthesis of DES and further transporting it to the gate of the factory [185]. The other one used for the boundary is the cradle-to-grave boundary, starting with the use phase of a product, its disposal and recycling, and ending at final disposal [184]. This will be the functional unit for these LCA studies as a basis for assessing the impacts of the system. It could be defined in terms of feedstock use, production of one bio-based product, or delivery of a number of bio-based products in a multifunctional system. Impact allocation among co-products is a problematic issue in biorefinery LCA studies, and this factor highly depends on it.

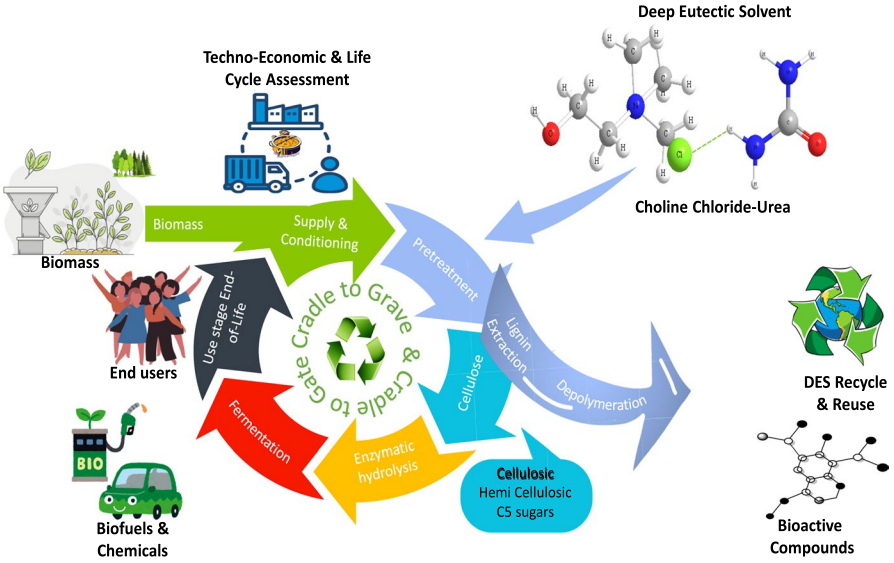


Fig. 8 The life cycle assessment process of biomass utilizes the DES-based pretreatment of the biomass reprinted from (Deep eutectic solvents for improved biomass pretreatment: Current status and future prospective towards sustainable processes https://doi.org/10.1016/j.biortech.2022.128396\) with permission from Elsevier.

Challenges and future perspectives in DESs

Due to the exceptional properties of DESs and their ability for efficient biomass deconstruction, they can be used in the pretreatment of lignocellulosic biomass; as a result, it has attracted great interest. Although the application of pretreating biomass using DES is beneficial, there are several challenges to fully supporting this method [187]. They can solubilize lignin and depolymerize the complex structure of biomass, providing more pores for enzymes or chemicals in downstream breaking stages [188]. DESs have significant potential to be applied to biomass pretreatment. Since they are excellent lignin solvents, DESs might be regarded as an alternative to conventional pretreatment methods and traditional organic cosolvent systems. They can remove lignin from biomass in a very selective manner while leaving the cellulose and hemicellulose fractions behind for enzymatic hydrolysis or chemical conversion. The solvation properties of DESs can be tuned by varying the eutectic mixture composition, making it possible to match solvent characteristics with diverse types of biomass [189].

A number of these avenues of future research have recently been explored in new papers. For instance, Ge et al. [28] employed explainable machine learning models together with Kamlet–Taft parameters to predict optimal DES formulations for high-efficiency lignin dissolution and provided a gateway to solvent design through AI. Wu et al. [29] synthesized supramolecular DES systems to enhance biomass adhesion and hydrolysis, illustrating how molecular structure can be engineered to optimize biomass-DES interaction. Liu et al. [83] in another publication used lignin solubility correlation with DES polarity parameters to screen the solvent. Additionally, Mamilla et al. [84] and Liang et al. [171] studied novel recycling processes, including membrane separation and anti-solvent recovery, to improve the efficiency and economy of the reutilization of DES. It is also future research that explores natural deep eutectic solvents (NADES), i.e., amino acid and sugar-based NADES, which offer higher biocompatibility for enzyme-based biorefineries [190, 191]. These innovations anticipate that breakthroughs from interdisciplinarity combining material science, biocatalysis, and digital technologies will be key for future innovations in biomass valorization based on DES.

The main problem of lignocellulosic biomass pretreatment with DESs is related to the lack of scale-up efforts; another concern could also be mass transfer. The viscosity of DESs, especially under given conditions that showcase a strong hydrogen bonding network among the constituents, may inhibit mass transfer through it into (e.g., reactants) and out from DES systems (enzymes, etc.). Furthermore, the high energy demand is one more drawback of DES-based pretreatment processes. Extensive energy usage in DES-based pretreatment comes primarily from the heating processes required to sustain operational temperature (~ 100 to 160 °C) and also in solvent recovery processes such as distillation or anti-solvent evaporation [192]. Moreover, DES's native viscosity might necessitate mechanical stirring or ultrasound/microwave-powered systems, again contributing to energy input [172, 193]. Research and development must continue to solve these challenges, to improve reactor design and mass transfer performance at the lowest levels of energy demand.

An additional concern is the co-extraction of biomass components when using DES media. This can be beneficial for recovering valuable compounds from biomass, but it could also have a rather negative effect on the effectiveness and recyclability of the DESs. We must address this co-extraction to optimize the production of biochar and design strategies to manage further downstream processing [194]. Furthermore, the moderate price of DESs in comparison with classic pretreatment solvents is one reason that restrains their expanded marketing. This can be combated by cost-effective recycling and reuse of DESs, which will further enhance the overall economic sustainability of this process [195].

Similarly, the compatibility of DESs with enzymes and microorganisms used in lignocellulosic biomass conversion processes should be addressed. These findings indicate that DESs may boost or decrease enzymatic activity, depending on the hydrophobicity and viscosity of their parts as well as the pH. Moreover, the effect of DESs on the viability and function of microbes needs to be determined, as this is essential in order to facilitate high biomass component conversion [196]. In conclusion, DES-based pretreatment of lignocellulosic biomass is a promising pathway toward improved efficiency and sustainability in the conversion processes of their structural polysaccharides. However, scale-up or mass transfer problems of the biofilm technology must be resolved before industrial application, and energy consumption, costs for mesh materials, including mechanical stability/toxicity/product inactivation during manufacturing, as well as compatibility with enzymes/microorganisms, become a limiting factor. Through these challenges, DESs are a potent solution in promoting sustainable biorefineries for biofuels, biochemicals, and other value-added products derived from lignocellulosic biomass [179, 197].

Summary

Although DESs present a great promise for biomass pretreatment, challenges like limited thermal and electrochemical stability, hygroscopicity, and reduced effectiveness with repeated use are some of the drawbacks of these solvents. For successful DES-based pretreatment, we herein optimize high-value bio-based product formation from lignocellulosic biomass. DESs' effect on pretreatment depends on several factors: the nature of the feedstock, component ratio, reaction temperature, and time. DESs have some significant benefits such as selective dissolution of lignin and hemicellulose, low toxicity, biodegradability, and simplicity of synthesis under mild conditions. They may also induce enzymatic hydrolysis and enhance sugar yield, and so they are well qualified for integration into biorefinery processes. Additionally, the tunability of DES formulation offers the potential for realizing specific applications according to biomass type and product of interest. Further studies are necessary to develop more efficient DES with specific selectivity for dissolution and separation of cellulose, hemicellulose, and lignin. This may be done by using molecular modeling and establishing the relationship between the structure and function of DES. Determining the physicochemical properties of DESs, including viscosity, polarity, and conductivity, will also be crucial to enhance pretreatment efficiency. Interactions

between reaction conditions, including temperature and solid/liquid ratio, will also need further elucidation for optimization in biomass processing. Moreover, one of the important issues that has to be overcome is the recyclability of DESs, which decreases after several runs. The combination of DESs with other techniques, such as microwave or ultrasound, may also constitute a new frontier in DES applications. Finally, life cycle assessments and economic analyses have to be considered in the development of DES-based biorefinery processes. Despite the challenges, DESs have great potential for the valorization of biomass sustainably, and further studies are necessary to fully develop their potential in biorefinery applications.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no competing interests.

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